Lab 8

Objective: Write a game in 6809
This lab builds on lectures 5, 6, and 7

Background
In this lab we will use what we learned in the previous lab to write a game in 6809 assembly.

Exercise 1: The Simulator's Keyboard
The simulator has a keyboard *memory mapped* to location $FF83. In this program we'll read a key from the keyboard and print it in the top left of the text screen. The keyboard is attached to a keyboard controller. To read a key we write $00 to $FF83 then read from $FF83. We get the key the user pressed, but if they haven't yet pressed one then we get $FF. Here's a routine that waits until the user presses a key and returns that key in A. See how it uses a loop.

```
keyboard EQU $FF83      ; The KEYBOARD is at memory location $FF83

GETCH:
  clr keyboard          ; write a 0 to the controller
  lda keyboard          ; get any key (if pressed)
  bmi GETCH             ; if the high bit was set then not valid
  rts
```

You already know where the text screen is located ($0400). Write the program that calls this routine and then writes the value of A to $0400.

a) Assemble and run your program.
b) ***Click on the text screen – keyboard input is only valid if it happens here***
b) Press a key.
c) The simulator is "retro". It only supports upper case.
d) Save your program as we're going to use it later.

Exercise 2: Guess the letter
This little game is guess the number, but with letters. It's a two player game so you can write it in pairs, or if you prefer, write it on your own. To write this you will probably use a loop, memory as variables, the simulator's keyboard, and output to the text screen.

The game progresses as follows. Player 1 presses a letter on the keyboard. The computer "remembers" that letter, player 2 then presses a letter on the keyboard. If they are the same letter then print a 'Y', otherwise, if the input letter comes before player 1's letter then print an 'A' (for "the hidden letter is ***after*** your guess") otherwise print 'B' (for "the hidden letter is ***before*** your guess"). If the answer is not correct (i.e. after, or before) then player 2 gets to try again and again until they guess the correct answer.